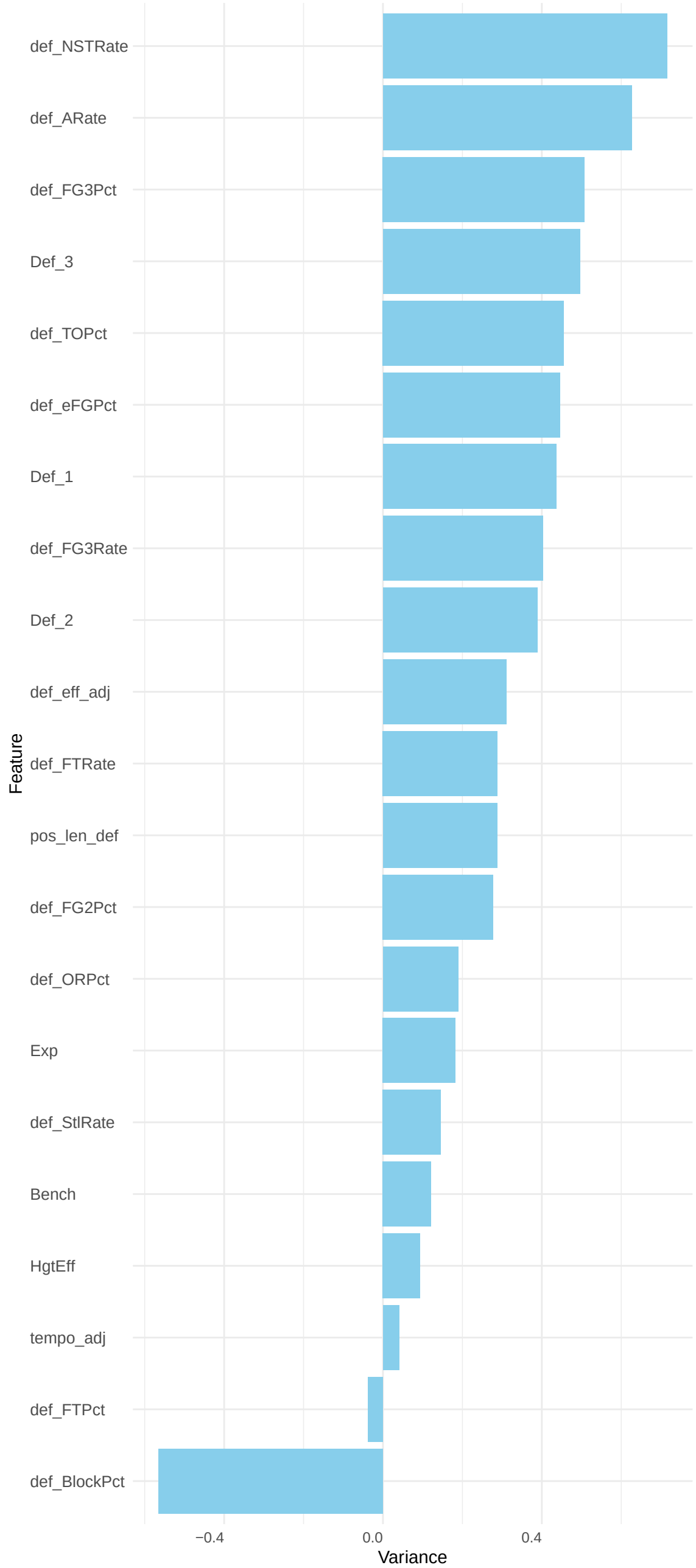
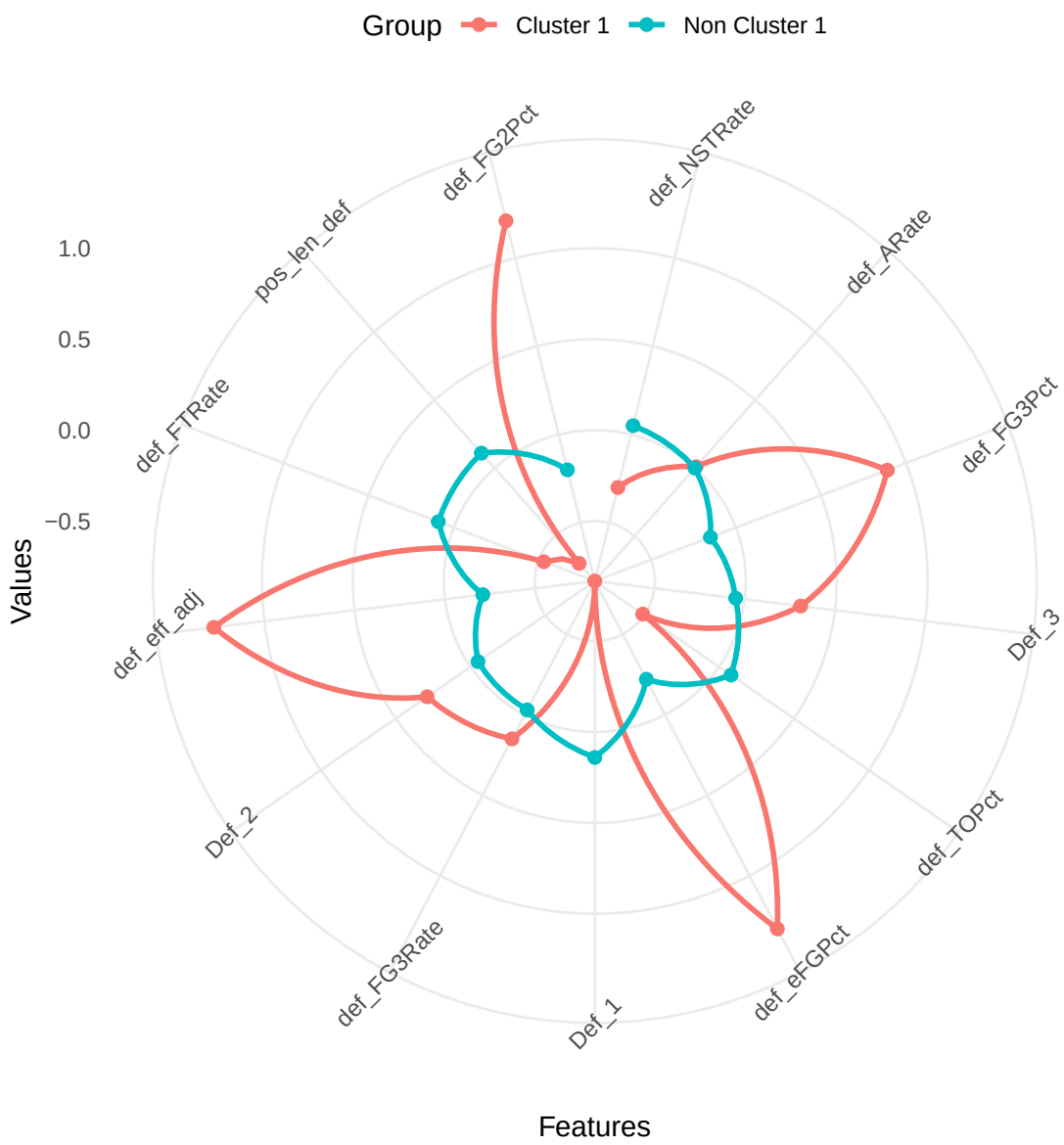
Cluster 1 − Variance Reduction and Polygon Plots

Feature Variance Decreases (Cluster 1)



Comparison of Cluster 1 vs Non Cluster 1 Features

Polygon Plot (Radar/Spider Chart)



Cluster 1 − Comparison Table

	Feature	Cluster 1	Non Cluster 1	Variance Reduction	Difference
1	def_NSTRate	−0.30	0.05	0.72	−0.35
2	def_ARate	0.01	0.00	0.63	0.01
3	def_FG3Pct	0.89	−0.15	0.51	1.03
4	Def_3	0.31	−0.05	0.50	0.36
5	def_TOPct	−0.51	0.08	0.46	−0.60
6	def_eFGPct	1.33	−0.22	0.45	1.55
7	Def_1	−0.83	0.14	0.44	−0.97
8	def_FG3Rate	0.15	−0.03	0.40	0.18
9	Def_2	0.29	−0.05	0.39	0.34
10	def_eff_adj	1.28	−0.21	0.31	1.49
11	def_FTRate	−0.53	0.09	0.29	−0.61
12	pos_len_def	−0.70	0.11	0.29	−0.81
13	def_FG2Pct	1.21	−0.20	0.28	1.41

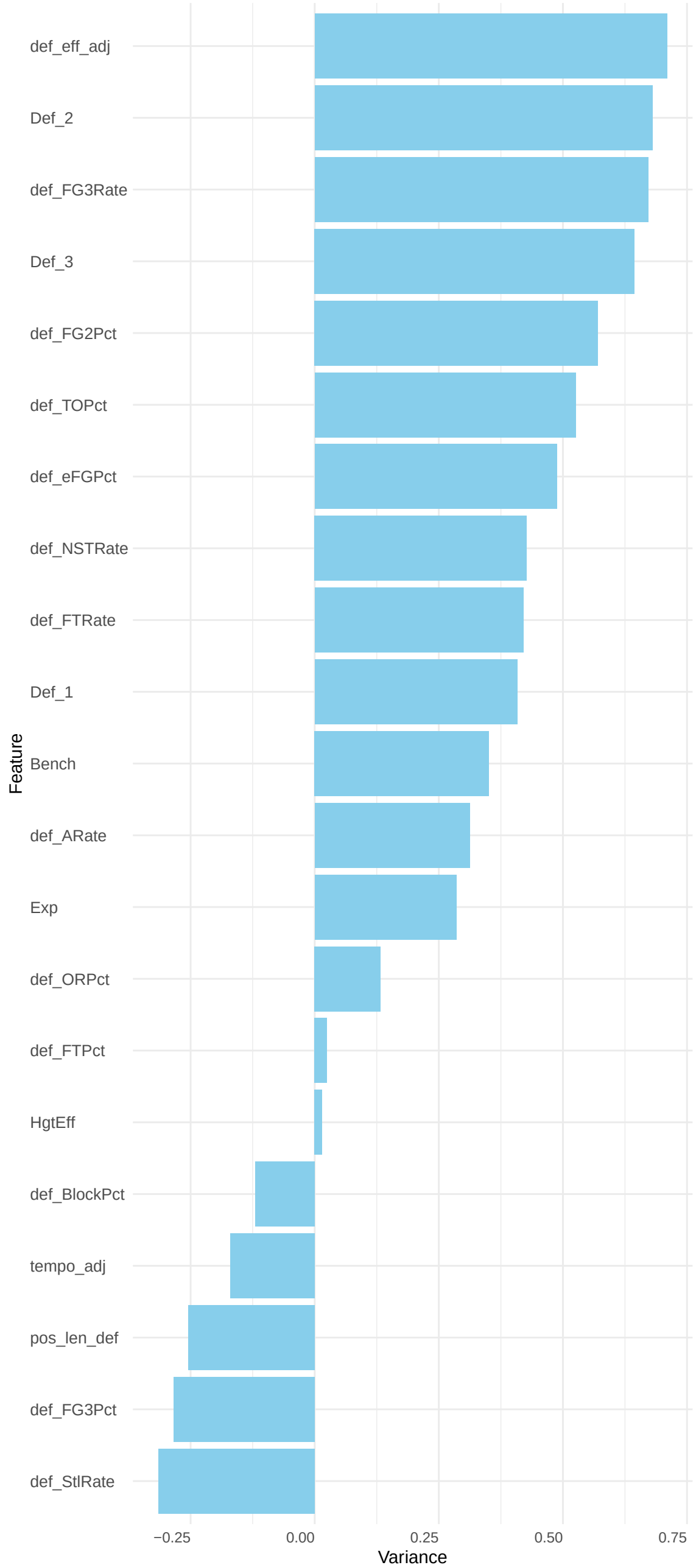
Cluster 1 − Tables Summary

	feature	p_value	t_stat
1	def_NSTRate	0.008	−2.75
2	def_ARate	0.924	0.10
3	def_FG3Pct	0.000	6.93
4	Def_3	0.024	2.33
5	def_TOPct	0.000	−3.80
6	def_eFGPct	0.000	10.15
7	Def_1	0.000	−6.17
8	def_FG3Rate	0.277	1.10
9	Def_2	0.044	2.08
10	def_eff_adj	0.000	8.91
11	def_FTRate	0.001	−3.53
12	pos_len_def	0.000	−4.70
13	def_FG2Pct	0.000	8.26
14	def_ORPct	0.278	1.10
15	Exp	0.112	−1.62
16	def_StlRate	0.061	1.93
17	Bench	0.763	0.30
18	HgtEff	0.247	1.18
19	tempo_adj	0.684	0.41

	TeamName	Season
1	California	2019
2	Miami FL	2019
3	Utah	2019
4	Wake Forest	2019
5	Washington St.	2019
6	Miami FL	2020
7	Nebraska	2020
8	Utah	2020
9	Boston College	2021
10	Miami FL	2021
11	TCU	2021
12	Wake Forest	2021
13	Georgia	2022
14	N.C. State	2022
15	Nebraska	2022
16	Oregon St.	2022
17	Stanford	2022
18	California	2023
19	LSU	2023
20	Louisville	2023
21	Minnesota	2023
22	South Carolina	2023
23	Wake Forest	2023
24	Louisville	2024
25	Michigan	2024
26	Stanford	2024
27	Syracuse	2024
28	Vanderbilt	2024
29	West Virginia	2024

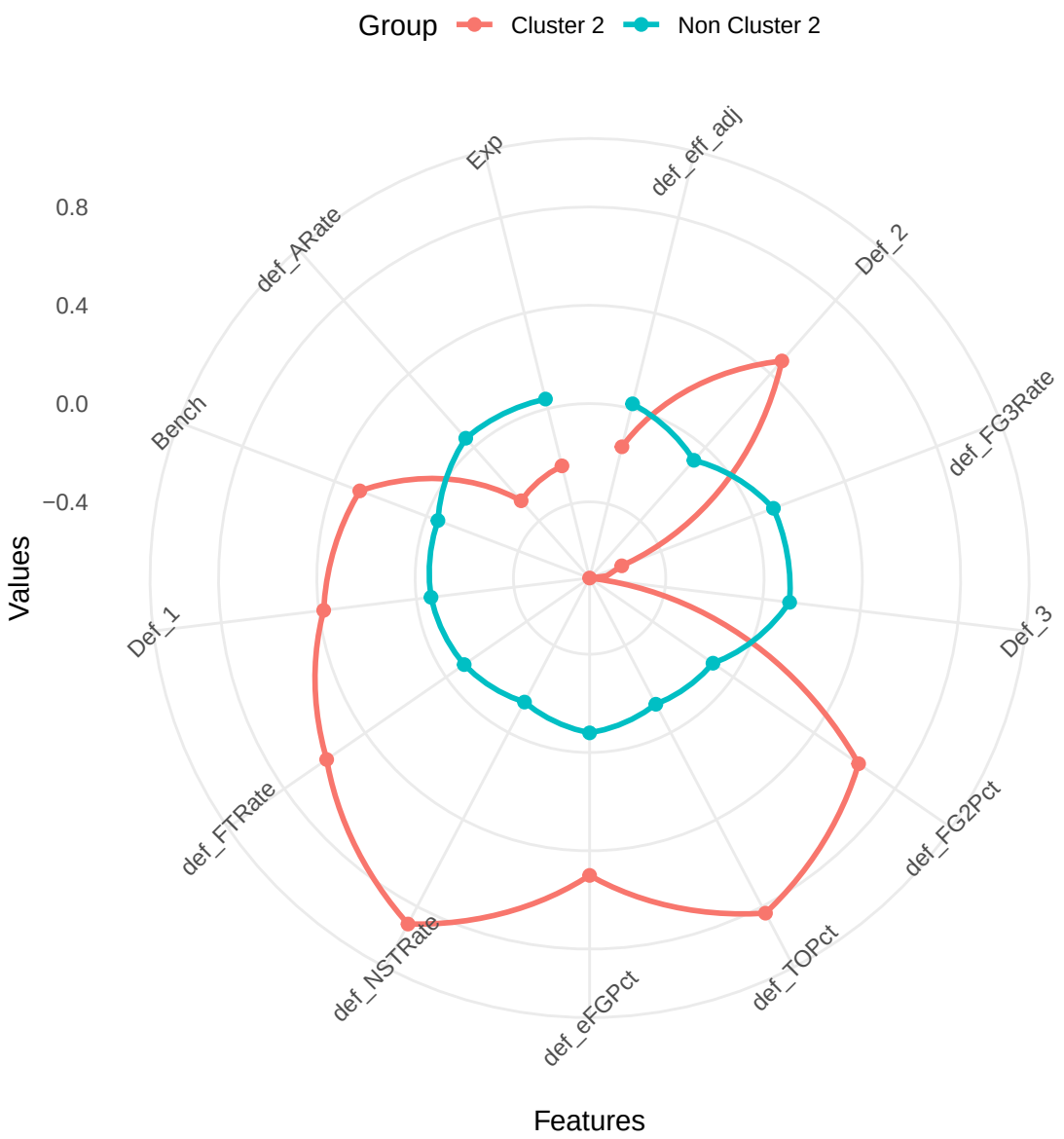
Cluster 2 − Variance Reduction and Polygon Plots

Feature Variance Decreases (Cluster 2)



Comparison of Cluster 2 vs Non Cluster 2 Features

Polygon Plot (Radar/Spider Chart)



Cluster 2 − Comparison Table

	Feature	Cluster 2	Non Cluster 2	Variance Reduction	Difference
1	def_eff_adj	−0.16	0.02	0.71	−0.18
2	Def_2	0.47	−0.07	0.68	0.55
3	def_FG3Rate	−0.57	0.09	0.67	−0.66
4	Def_3	−0.71	0.11	0.64	−0.82
5	def_FG2Pct	0.62	−0.10	0.57	0.71
6	def_TOPct	0.83	−0.13	0.53	0.97
7	def_eFGPct	0.50	−0.08	0.49	0.58
8	def_NSTRate	0.88	−0.14	0.43	1.02
9	def_FTRate	0.59	−0.09	0.42	0.68
10	Def_1	0.38	−0.06	0.41	0.44
11	Bench	0.29	−0.05	0.35	0.34
12	def_ARate	−0.29	0.05	0.31	−0.33
13	Exp	−0.24	0.04	0.29	−0.28

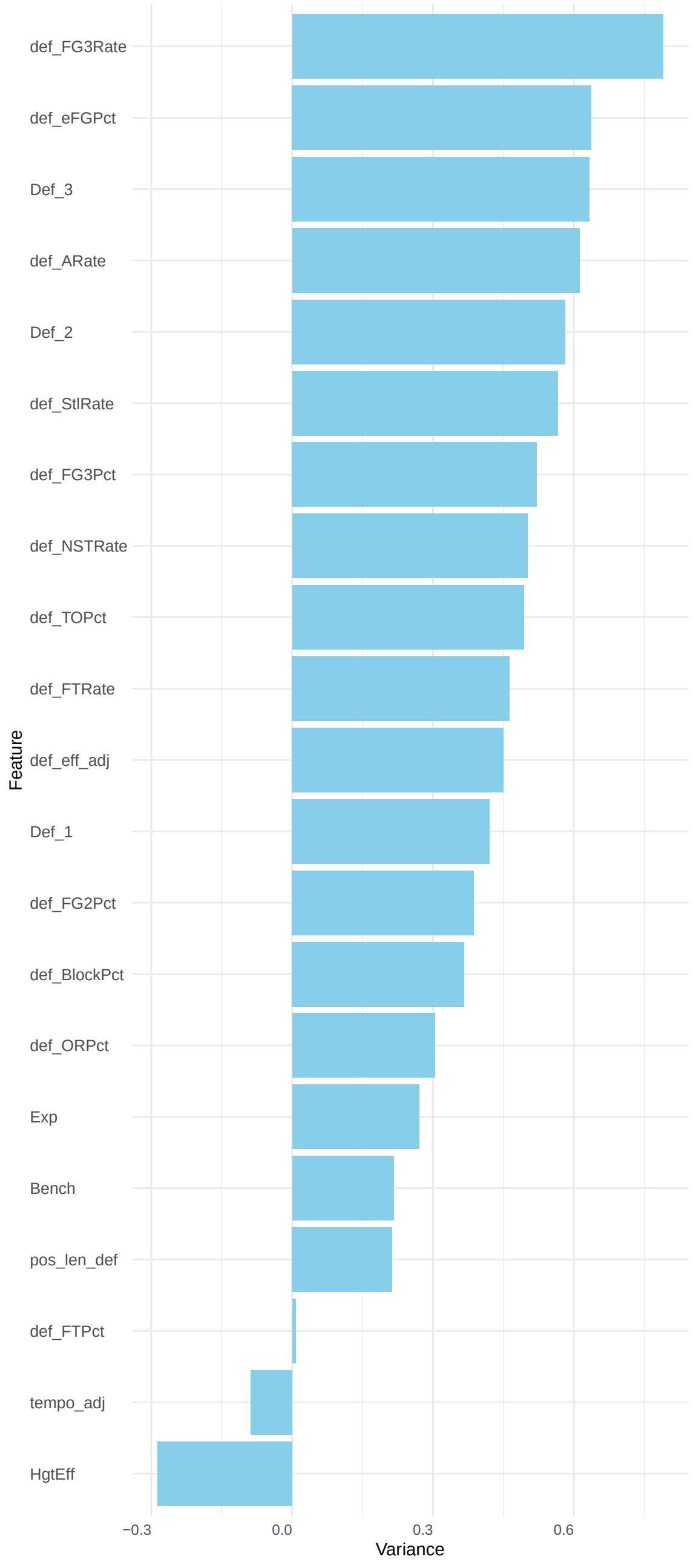
Cluster 2 − Tables Summary

	feature	p_value	t_stat
1	def_eff_adj	0.163	−1.41
2	Def_2	0.000	4.16
3	def_FG3Rate	0.000	−4.95
4	Def_3	0.000	−6.06
5	def_FG2Pct	0.000	4.91
6	def_TOPct	0.000	6.47
7	def_eFGPct	0.001	3.74
8	def_NSTRate	0.000	6.39
9	def_FTRate	0.000	4.19
10	Def_1	0.011	2.65
11	Bench	0.054	1.98
12	def_ARate	0.063	−1.91
13	Exp	0.126	−1.56
14	def_ORPct	0.026	2.31
15	def_FTPct	0.515	−0.66
16	HgtEff	0.000	−4.05

	TeamName	Season
1	Arizona	2019
2	Illinois	2019
3	Vanderbilt	2019
4	West Virginia	2019
5	Alabama	2020
6	Arkansas	2020
7	Boston College	2020
8	Georgia	2020
9	Iowa St.	2020
10	TCU	2020
11	Vanderbilt	2020
12	Washington St.	2020
13	Arizona St.	2021
14	Auburn	2021
15	California	2021
16	Duke	2021
17	Georgia	2021
18	Iowa St.	2021
19	Minnesota	2021
20	N.C. State	2021
21	Penn St.	2021
22	South Carolina	2021
23	Utah	2021
24	Vanderbilt	2021
25	Kansas St.	2022
26	Vanderbilt	2022
27	Washington	2022
28	West Virginia	2022

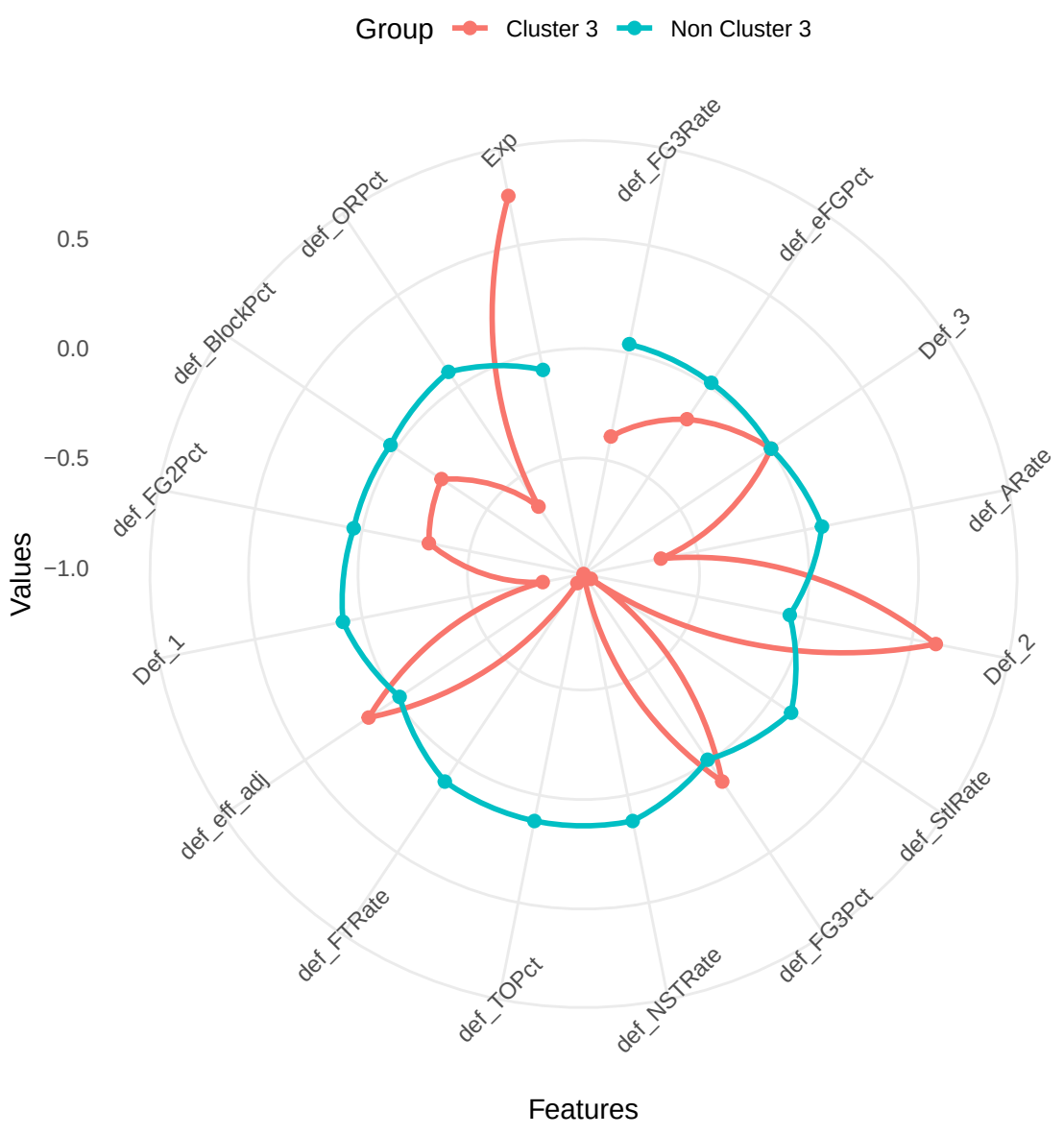
Cluster 3 − Variance Reduction and Polygon Plots

Feature Variance Decreases (Cluster 3)



Comparison of Cluster 3 vs Non Cluster 3 Features

Polygon Plot (Radar/Spider Chart)



Cluster 3 − Comparison Table

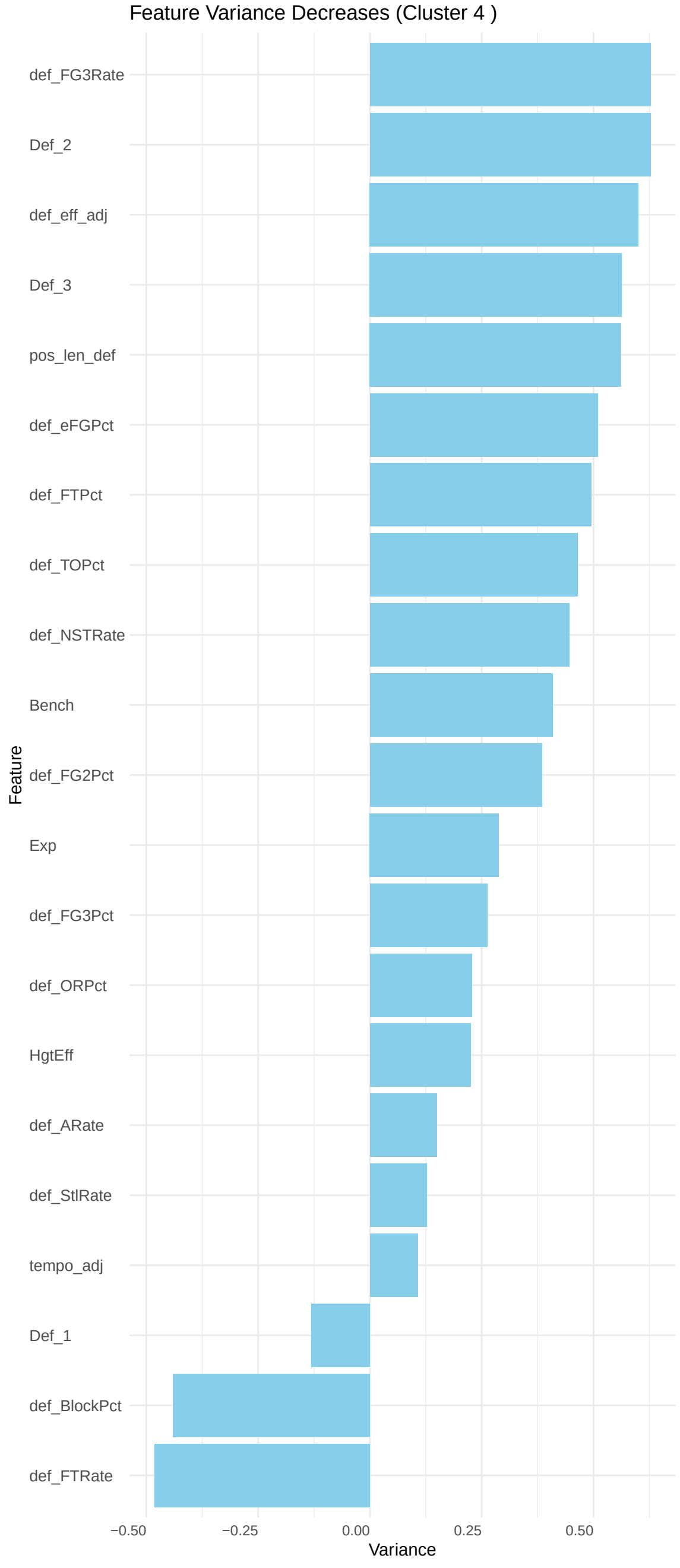
	Feature	Cluster 3	Non Cluster 3	Variance Reduction	Difference
1	def_FG3Rate	−0.39	0.04	0.79	−0.44
2	def_eFGPct	−0.18	0.02	0.64	−0.20
3	Def_3	0.00	0.00	0.63	0.00
4	def_ARate	−0.67	0.08	0.61	−0.74
5	Def_2	0.61	−0.07	0.58	0.68
6	def_StlRate	−0.99	0.11	0.57	−1.11
7	def_FG3Pct	0.11	−0.01	0.52	0.12
8	def_NSTRate	−1.01	0.12	0.50	−1.13
9	def_TOPct	−1.03	0.12	0.49	−1.15
10	def_FTRate	−0.98	0.11	0.46	−1.09
11	def_eff_adj	0.15	−0.02	0.45	0.16
12	Def_1	−0.84	0.09	0.42	−0.93
13	def_FG2Pct	−0.31	0.04	0.39	−0.35
14	def_BlockPct	−0.25	0.03	0.37	−0.28
15	def_ORPct	−0.66	0.08	0.30	−0.74
16	Exp	0.73	−0.08	0.27	0.82

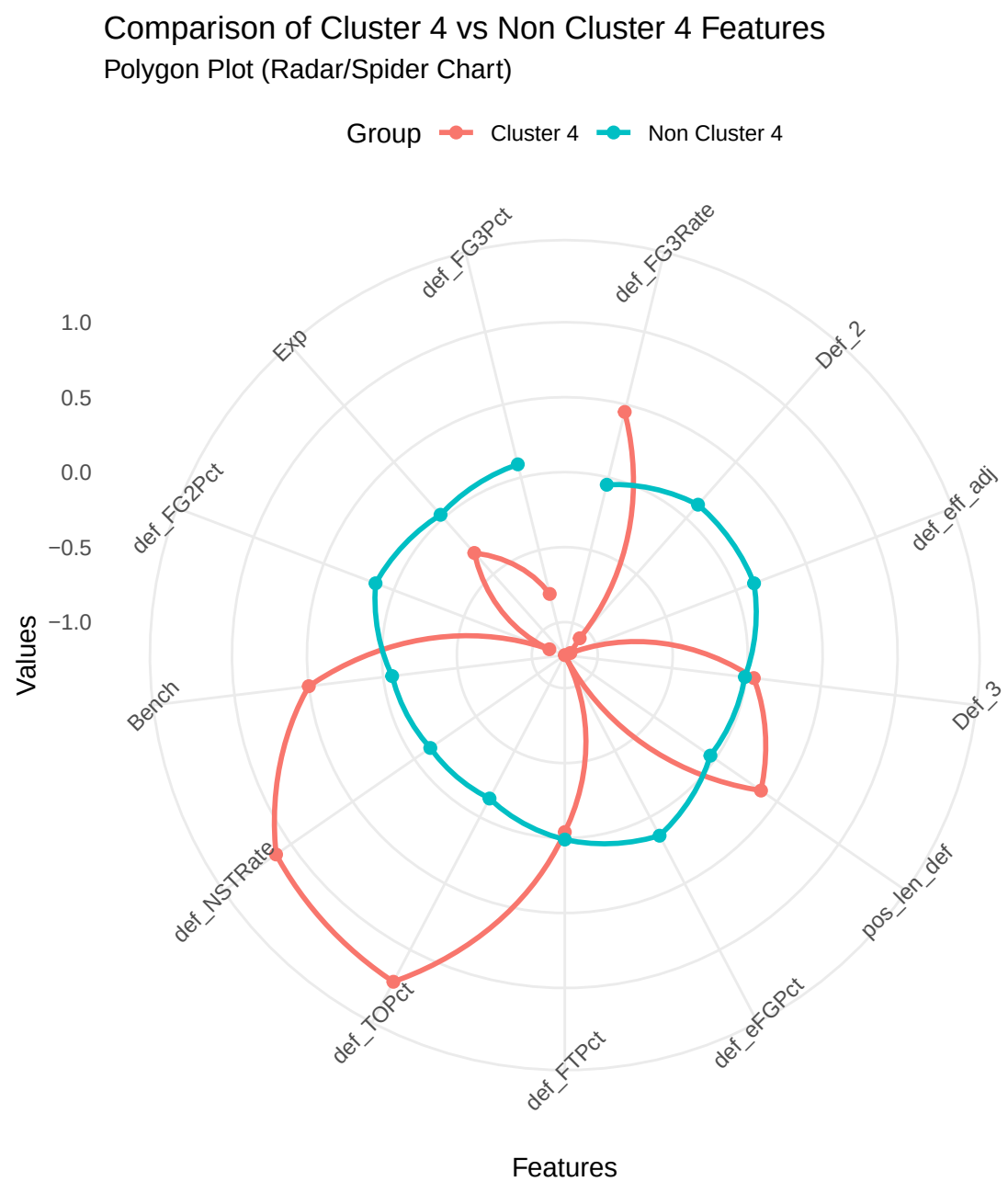
Cluster 3 − Tables Summary

	feature	p_value	t_stat
1	def_FG3Rate	0.001	−3.47
2	def_eFGPct	0.204	−1.29
3	Def_3	0.981	0.02
4	def_ARate	0.000	−4.79
5	Def_2	0.000	4.29
6	def_StlRate	0.000	−6.90
7	def_FG3Pct	0.472	0.73
8	def_NSTRate	0.000	−6.66
9	def_TOPct	0.000	−6.72
10	def_FTRate	0.000	−6.22
11	def_eff_adj	0.365	0.92
12	Def_1	0.000	−5.14
13	def_FG2Pct	0.074	−1.86
14	def_BlockPct	0.151	−1.48
15	def_ORPct	0.001	−3.75
16	Exp	0.000	4.09
17	Bench	0.000	−6.18
18	pos_len_def	0.016	2.59
19	def_FTPct	0.264	1.14

	TeamName	Season
1	Nebraska	2019
2	Notre Dame	2019
3	Notre Dame	2020
4	Notre Dame	2021
5	Maryland	2022
6	Minnesota	2022
7	Penn St.	2022
8	Virginia	2022
9	Michigan	2023
10	North Carolina	2023
11	Notre Dame	2023
12	Ohio St.	2023
13	Oregon	2023
14	Virginia Tech	2023
15	Wisconsin	2023
16	Boston College	2024
17	California	2024
18	Ohio St.	2024
19	Pittsburgh	2024
20	Utah	2024
21	Wake Forest	2024

Cluster 4 − Variance Reduction and Polygon Plots





Cluster 4 − Comparison Table

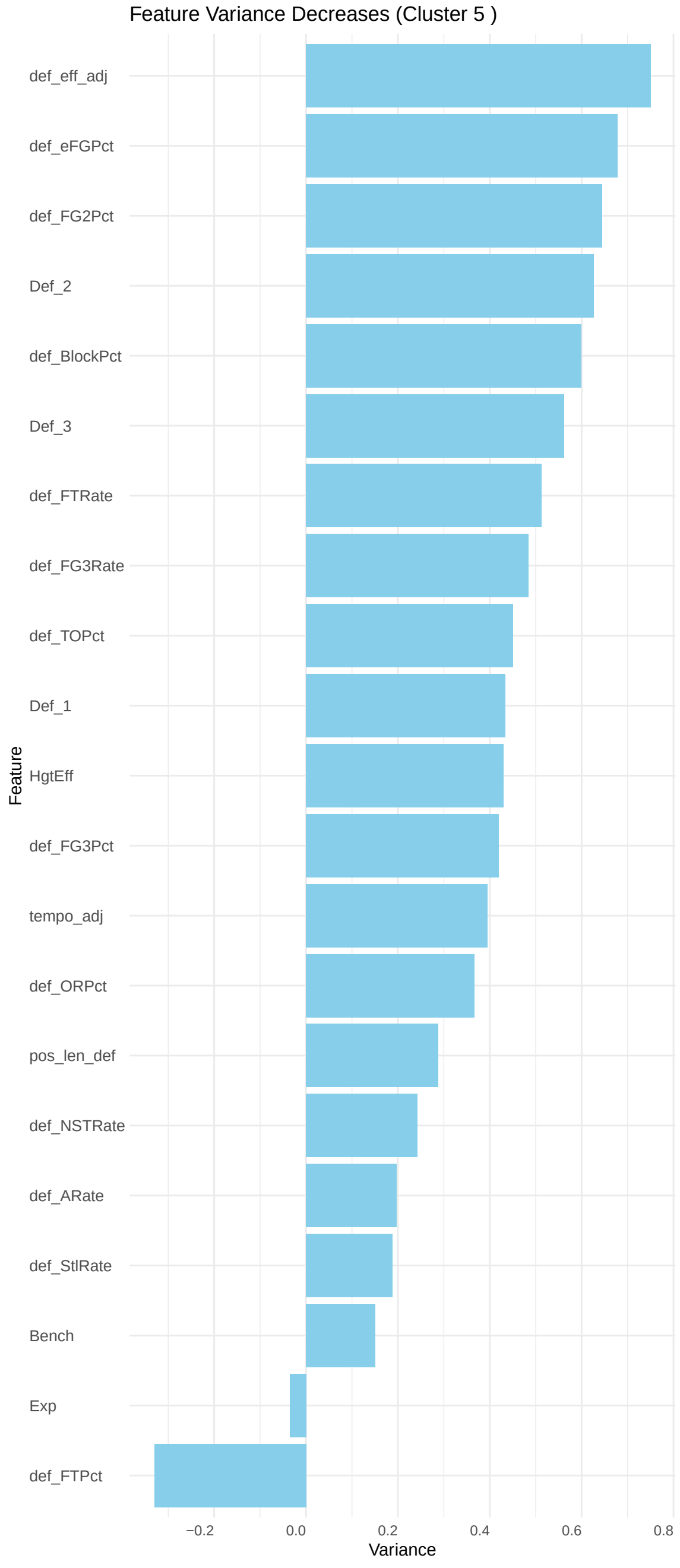
	Feature	Cluster 4	Non Cluster 4	Variance Reduction	Difference
1	def_FG3Rate	0.45	−0.05	0.63	0.50
2	Def_2	−1.07	0.12	0.63	−1.20
3	def_eff_adj	−1.18	0.13	0.60	−1.32
4	Def_3	0.05	−0.01	0.56	0.05
5	pos_len_def	0.37	−0.04	0.56	0.41
6	def_eFGPct	−1.22	0.14	0.51	−1.36
7	def_FTPct	−0.04	0.01	0.49	−0.05
8	def_TOPct	1.24	−0.14	0.46	1.38
9	def_NSTRate	1.12	−0.13	0.45	1.25
10	Bench	0.50	−0.06	0.41	0.55
11	def_FG2Pct	−1.11	0.13	0.38	−1.24
12	Exp	−0.31	0.03	0.29	−0.34
13	def_FG3Pct	−0.80	0.09	0.26	−0.89

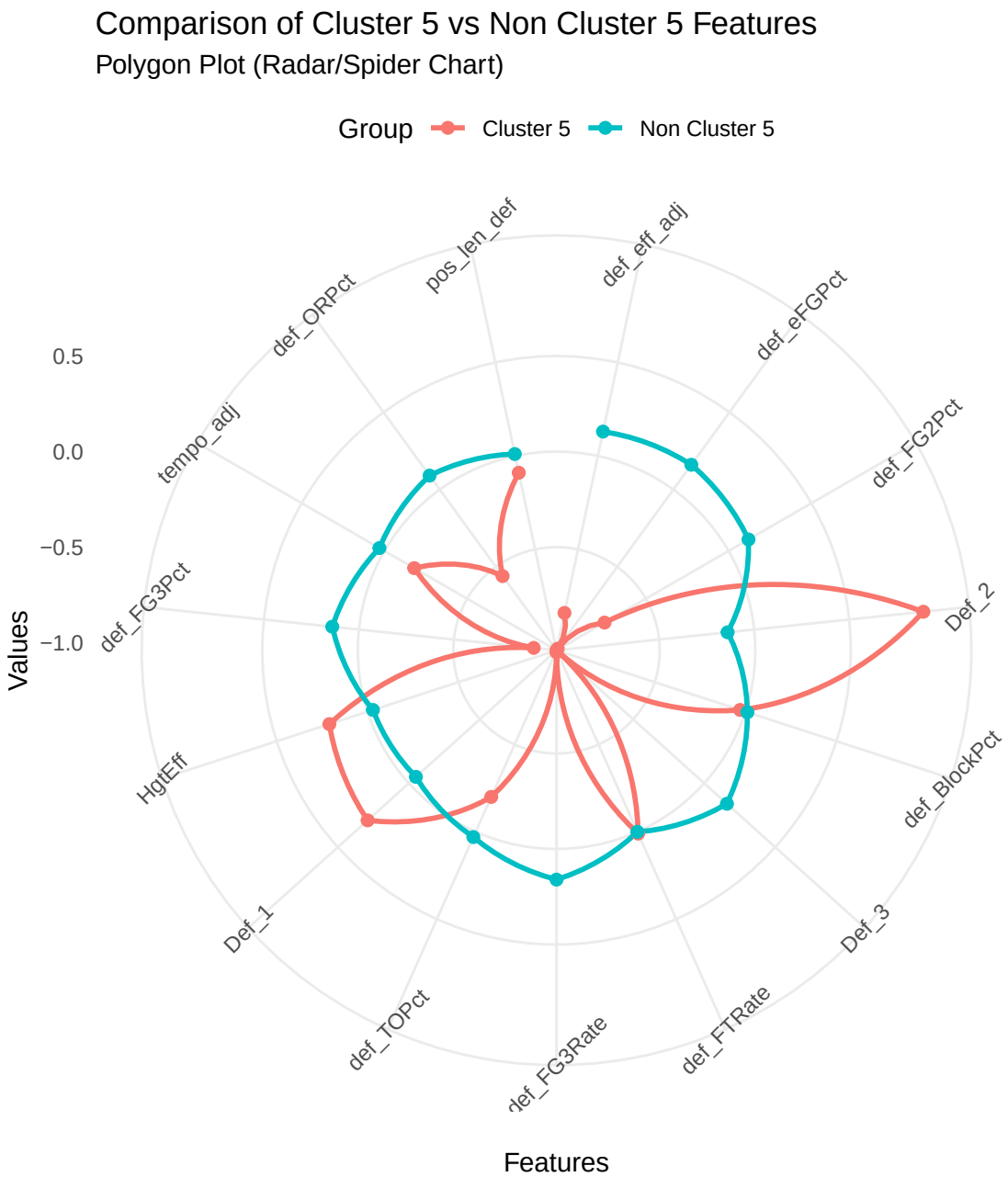
Cluster 4 − Tables Summary

	feature	p_value	t_stat
1	def_FG3Rate	0.002	3.28
2	Def_2	0.000	−7.93
3	def_eff_adj	0.000	−8.52
4	Def_3	0.747	0.32
5	pos_len_def	0.017	2.52
6	def_eFGPct	0.000	−8.09
7	def_FTPct	0.777	−0.29
8	def_TOPct	0.000	7.97
9	def_NSTRate	0.000	7.05
10	Bench	0.005	3.01
11	def_FG2Pct	0.000	−6.71
12	Exp	0.096	−1.73
13	def_FG3Pct	0.000	−4.43
14	def_ORPct	0.048	2.08
15	HgtEff	0.106	−1.67
16	def_ARate	0.056	2.01
17	def_StlRate	0.037	2.20
18	tempo_adj	0.671	0.43

	TeamName	Season
1	Arkansas	2019
2	Georgia Tech	2019
3	Missouri	2019
4	Pittsburgh	2019
5	South Carolina	2019
6	Kansas St.	2020
7	Missouri	2020
8	Oklahoma St.	2020
9	Purdue	2020
10	South Carolina	2020
11	Stanford	2020
12	Tennessee	2020
13	Washington	2020
14	Mississippi	2021
15	Arizona St.	2022
16	Oklahoma St.	2022
17	South Carolina	2022
18	Texas A&M	2022
19	Oklahoma St.	2023
20	Rutgers	2024
21	UCF	2024

Cluster 5 − Variance Reduction and Polygon Plots





Cluster 5 − Comparison Table

	Feature	Cluster 5	Non Cluster 5	Variance Reduction	Difference
1	def_eff_adj	−0.84	0.13	0.75	−0.97
2	def_eFGPct	−1.03	0.16	0.68	−1.19
3	def_FG2Pct	−0.75	0.12	0.64	−0.87
4	Def_2	0.89	−0.14	0.63	1.03
5	def_BlockPct	−0.03	0.01	0.60	−0.04
6	Def_3	−1.04	0.16	0.56	−1.21
7	def_FTRate	0.01	0.00	0.51	0.01
8	def_FG3Rate	−1.03	0.16	0.48	−1.19
9	def_TOPct	−0.20	0.03	0.45	−0.24
10	Def_1	0.29	−0.05	0.43	0.34
11	HgtEff	0.21	−0.03	0.43	0.25
12	def_FG3Pct	−0.92	0.14	0.42	−1.06
13	tempo_adj	−0.18	0.03	0.39	−0.21
14	def_ORPct	−0.56	0.09	0.37	−0.65
15	pos_len_def	−0.09	0.01	0.29	−0.10

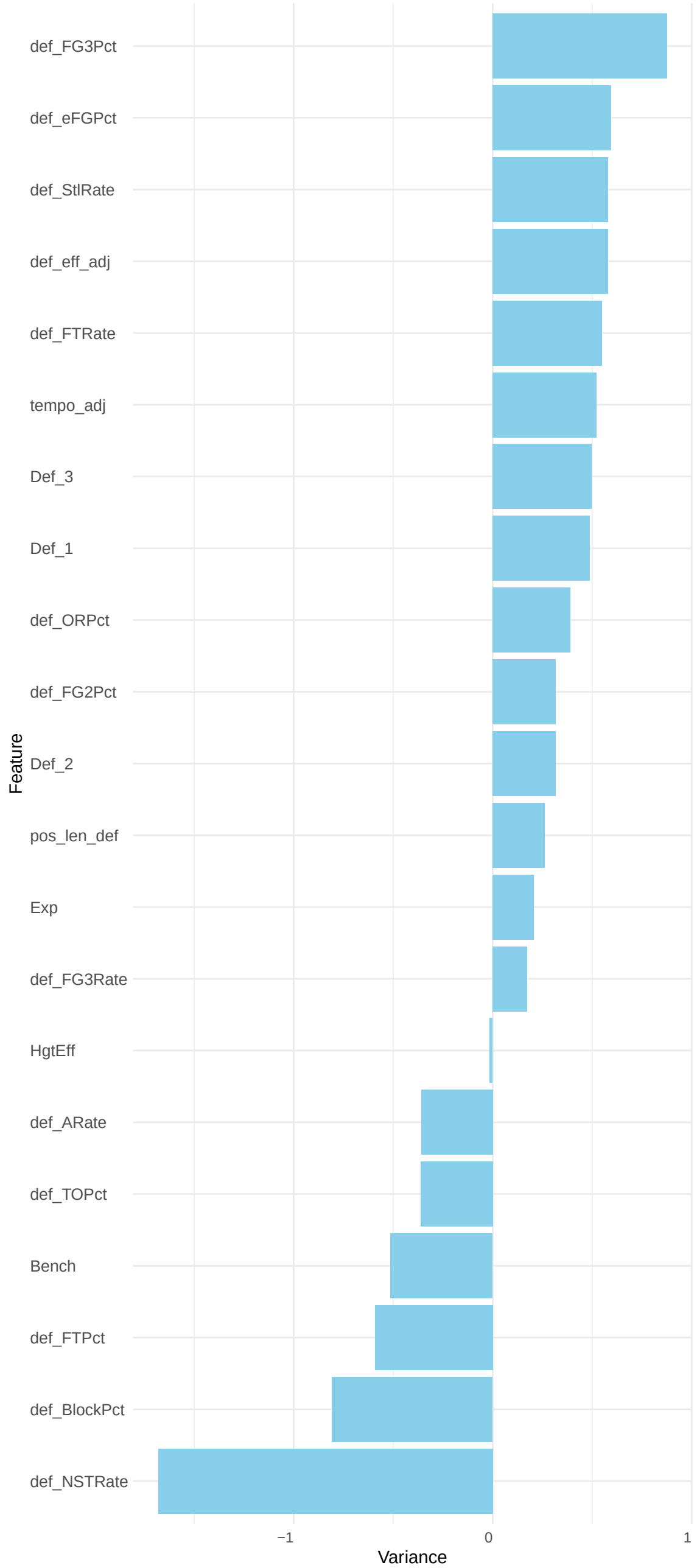
Cluster 5 − Tables Summary

	feature	p_value	t_stat
1	def_eff_adj	0.000	−8.04
2	def_eFGPct	0.000	−9.20
3	def_FG2Pct	0.000	−6.44
4	Def_2	0.000	7.54
5	def_BlockPct	0.792	−0.26
6	Def_3	0.000	−8.40
7	def_FTRate	0.964	0.05
8	def_FG3Rate	0.000	−7.80
9	def_TOPct	0.149	−1.47
10	Def_1	0.043	2.08
11	HgtEff	0.133	1.53
12	def_FG3Pct	0.000	−6.59
13	tempo_adj	0.224	−1.23
14	def_ORPct	0.000	−3.87
15	pos_len_def	0.577	−0.56
16	def_NSTRate	0.800	−0.26
17	def_ARate	0.000	−4.18
18	def_StlRate	0.101	1.68
19	Bench	0.193	−1.33

	TeamName	Season
1	Alabama	2019
2	Colorado	2019
3	Northwestern	2019
4	Penn St.	2019
5	Stanford	2019
6	TCU	2019
7	Georgia Tech	2020
8	Minnesota	2020
9	Mississippi	2020
10	Mississippi St.	2020
11	Texas	2020
12	Kentucky	2021
13	Washington St.	2021
14	California	2022
15	Colorado	2022
16	Oklahoma	2022
17	Wake Forest	2022
18	Washington St.	2022
19	Colorado	2023
20	Florida	2023
21	Oklahoma	2023
22	Utah	2023
23	Washington St.	2023
24	Cincinnati	2024
25	Kansas St.	2024
26	Maryland	2024
27	Notre Dame	2024
28	Oklahoma	2024

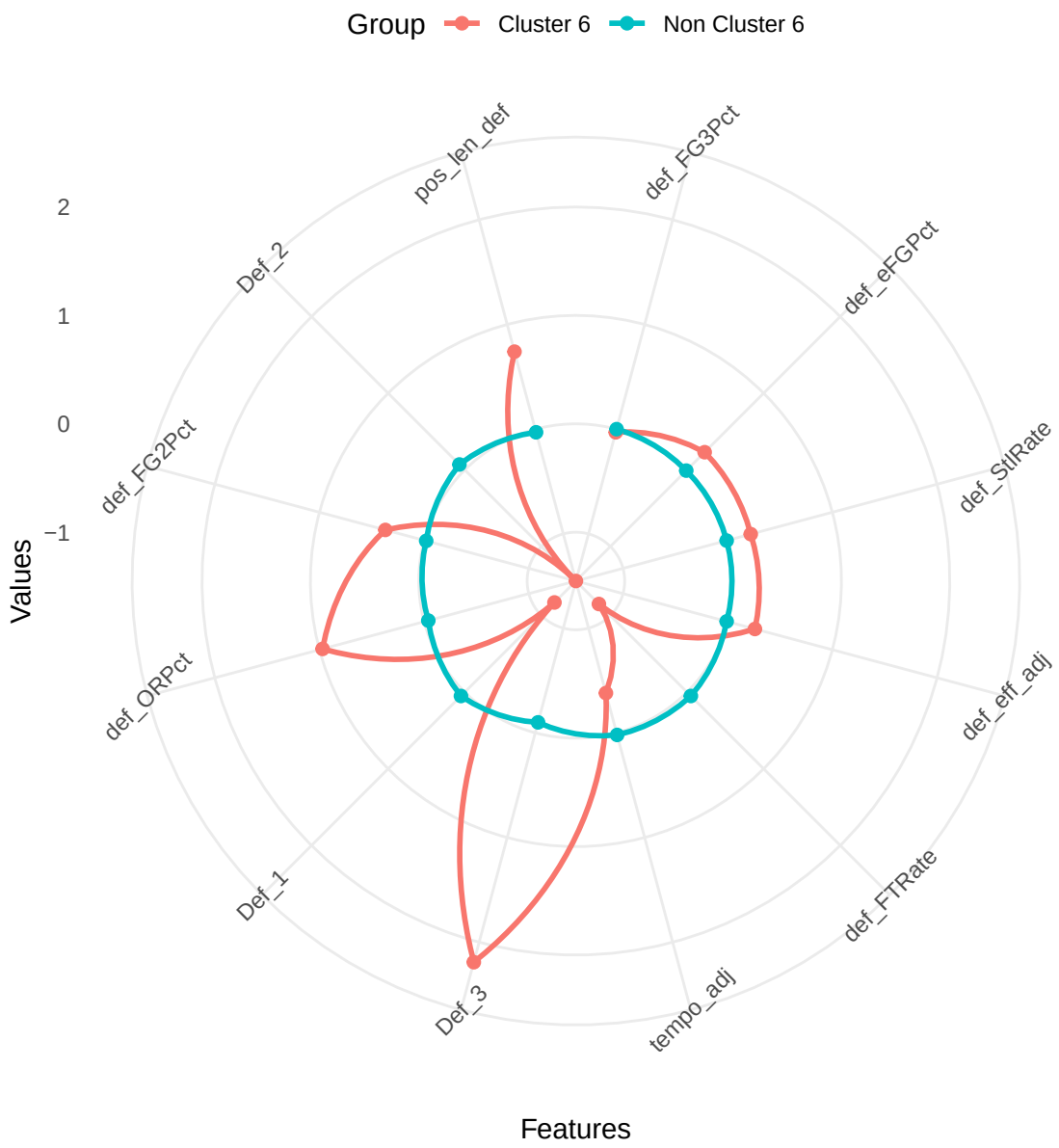
Cluster 6 − Variance Reduction and Polygon Plots

Feature Variance Decreases (Cluster 6)



Comparison of Cluster 6 vs Non Cluster 6 Features

Polygon Plot (Radar/Spider Chart)



Cluster 6 − Comparison Table

	Feature	Cluster 6	Non Cluster 6	Variance Reduction	Difference
1	def_FG3Pct	−0.03	0.00	0.88	−0.03
2	def_eFGPct	0.23	−0.01	0.60	0.24
3	def_StlRate	0.22	−0.01	0.58	0.23
4	def_eff_adj	0.26	−0.01	0.58	0.27
5	def_FTRate	−1.15	0.05	0.55	−1.20
6	tempo_adj	−0.38	0.02	0.52	−0.39
7	Def_3	2.19	−0.10	0.50	2.29
8	Def_1	−1.17	0.05	0.49	−1.23
9	def_ORPct	0.97	−0.04	0.39	1.01
10	def_FG2Pct	0.37	−0.02	0.32	0.39
11	Def_2	−1.45	0.07	0.32	−1.51
12	pos_len_def	0.74	−0.03	0.26	0.77

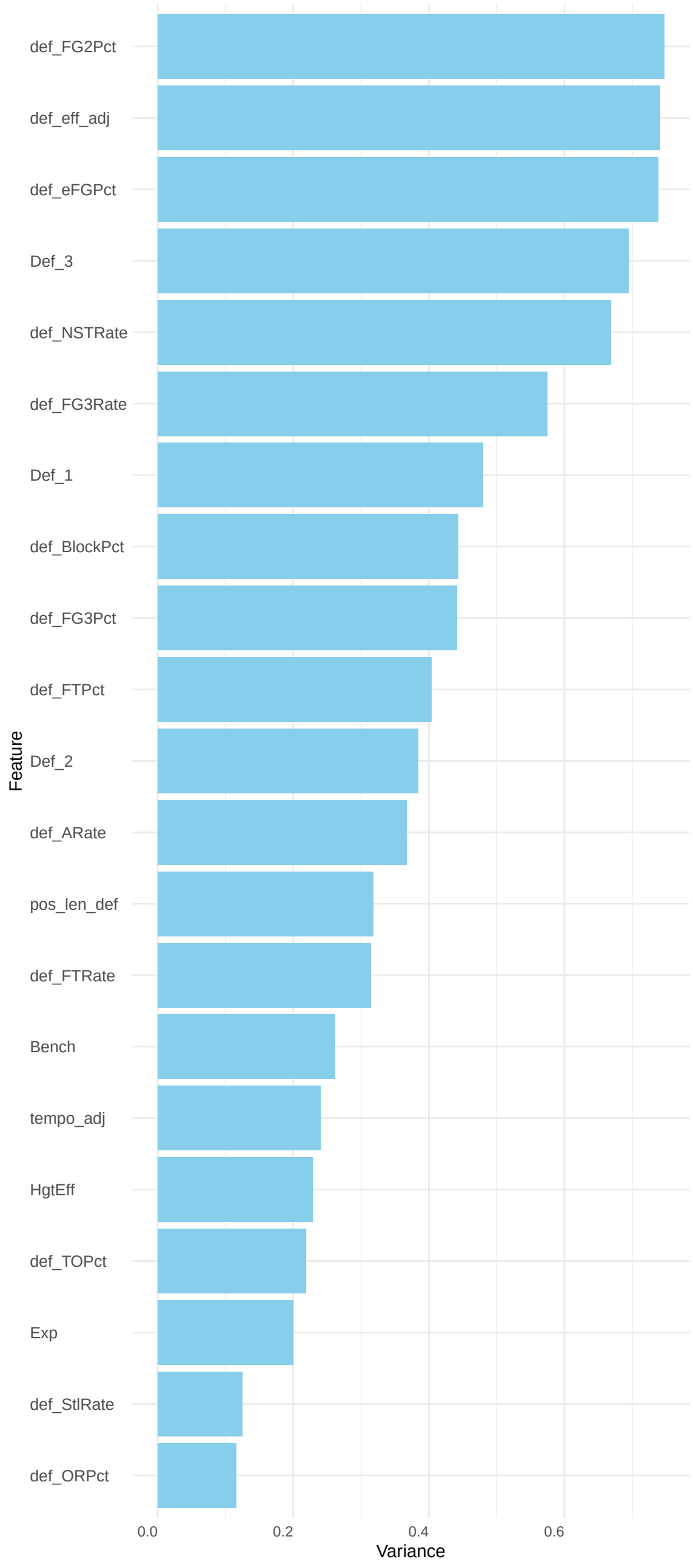
Cluster 6 − Tables Summary

	feature	p_value	t_stat
1	def_FG3Pct	0.817	−0.23
2	def_eFGPct	0.313	1.06
3	def_StlRate	0.332	1.02
4	def_eff_adj	0.259	1.20
5	def_FTRate	0.001	−5.13
6	tempo_adj	0.135	−1.63
7	Def_3	0.000	9.33
8	Def_1	0.001	−4.92
9	def_ORPct	0.004	3.75
10	def_FG2Pct	0.203	1.37
11	Def_2	0.000	−5.33
12	pos_len_def	0.028	2.62
13	Exp	0.645	−0.48
14	def_FG3Rate	0.000	7.67

	TeamName	Season
1	Texas A&M	2019
2	Pittsburgh	2020
3	Syracuse	2020
4	Texas A&M	2020
5	Texas A&M	2021
6	Syracuse	2022
7	Nebraska	2023
8	Syracuse	2023
9	Miami FL	2024

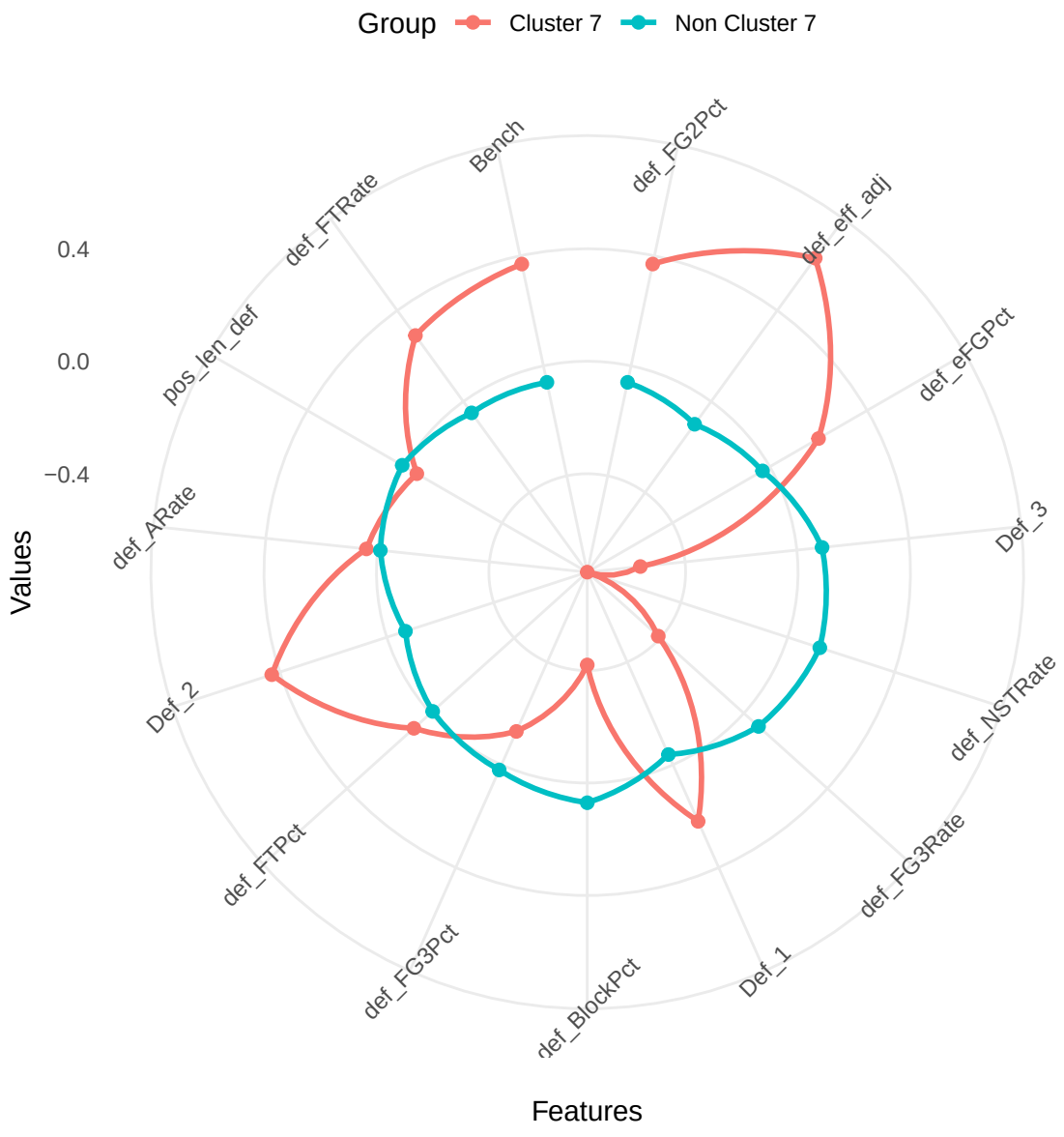
Cluster 7 − Variance Reduction and Polygon Plots

Feature Variance Decreases (Cluster 7)



Comparison of Cluster 7 vs Non Cluster 7 Features

Polygon Plot (Radar/Spider Chart)



Cluster 7 − Comparison Table

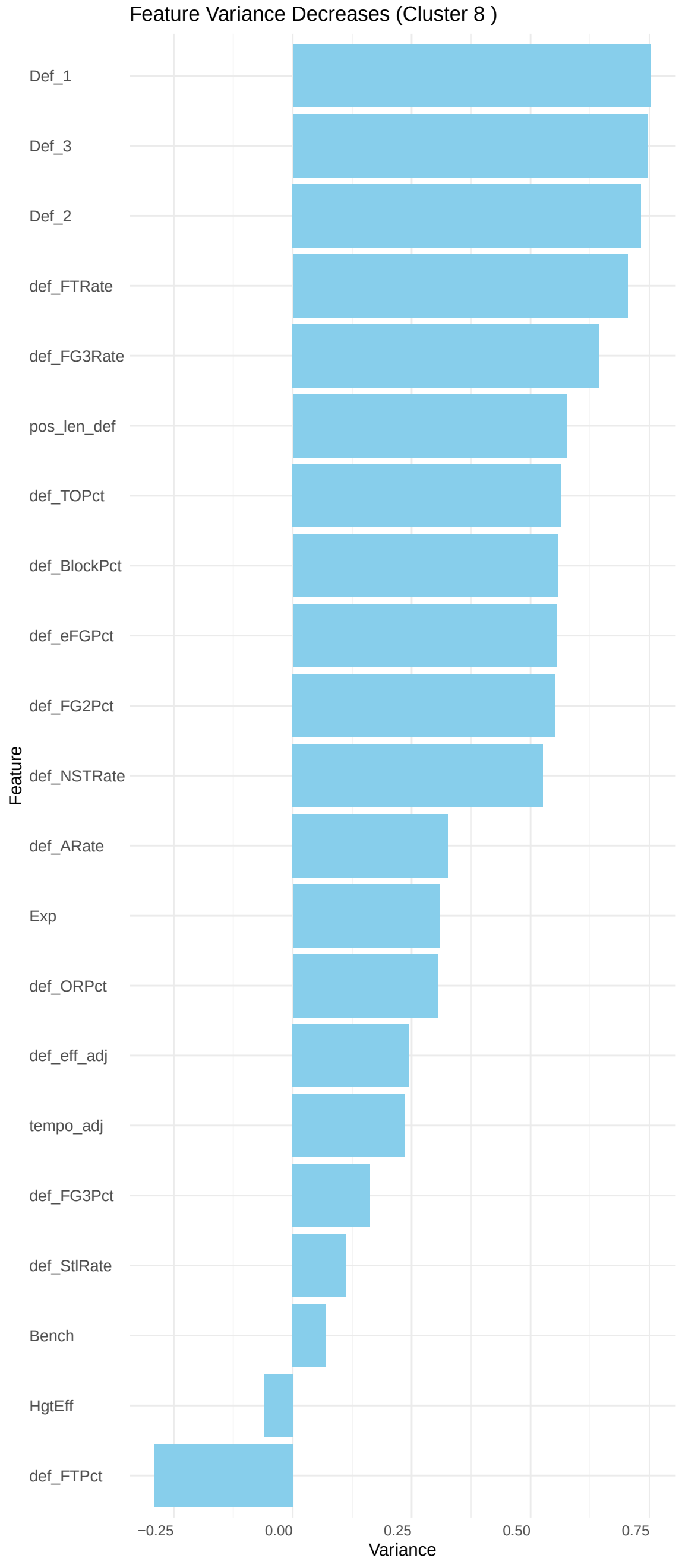
	Feature	Cluster 7	Non Cluster 7	Variance Reduction	Difference
1	def_FG2Pct	0.37	−0.06	0.75	0.43
2	def_eff_adj	0.63	−0.10	0.74	0.73
3	def_eFGPct	0.20	−0.03	0.74	0.24
4	Def_3	−0.56	0.09	0.69	−0.65
5	def_NSTRate	−0.75	0.12	0.67	−0.88
6	def_FG3Rate	−0.41	0.07	0.57	−0.48
7	Def_1	0.22	−0.04	0.48	0.26
8	def_BlockPct	−0.42	0.07	0.44	−0.49
9	def_FG3Pct	−0.13	0.02	0.44	−0.15
10	def_FTPct	0.08	−0.01	0.40	0.09
11	Def_2	0.43	−0.07	0.38	0.50
12	def_ARate	0.04	−0.01	0.37	0.05
13	pos_len_def	−0.05	0.01	0.32	−0.06
14	def_FTRate	0.29	−0.05	0.31	0.34
15	Bench	0.37	−0.06	0.26	0.43

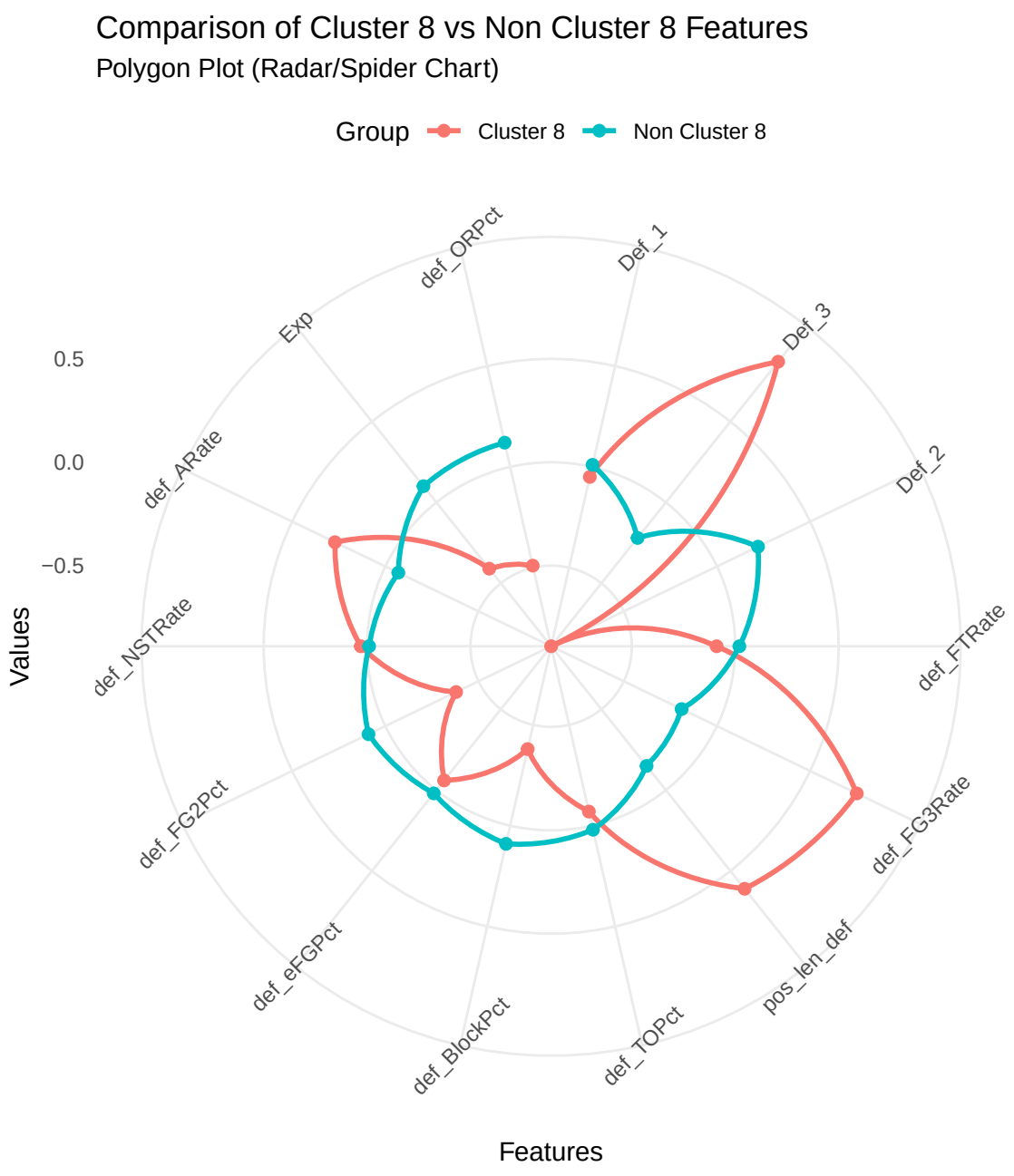
Cluster 7 − Tables Summary

	feature	p_value	t_stat
1	def_FG2Pct	0.001	3.55
2	def_eff_adj	0.000	6.00
3	def_eFGPct	0.058	1.93
4	Def_3	0.000	−5.08
5	def_NSTRate	0.000	−6.70
6	def_FG3Rate	0.001	−3.35
7	Def_1	0.103	1.66
8	def_BlockPct	0.003	−3.12
9	def_FG3Pct	0.353	−0.94
10	def_FTPct	0.577	0.56
11	Def_2	0.004	3.05
12	def_ARate	0.776	0.29
13	pos_len_def	0.740	−0.33
14	def_FTRate	0.056	1.97
15	Bench	0.020	2.42
16	tempo_adj	0.154	1.45
17	HgtEff	0.538	0.62
18	def_TOPct	0.027	−2.30
19	Exp	0.000	6.03
20	def_StlRate	0.230	1.22
21	def_ORPct	0.001	3.68

	TeamName	Season
1	Georgia	2019
2	Wake Forest	2020
3	Washington	2021
4	Florida	2022
5	Georgia Tech	2022
6	Mississippi	2022
7	Missouri	2022
8	Oregon	2022
9	Utah	2022
10	Georgia	2023
11	Georgia Tech	2023
12	Mississippi	2023
13	Vanderbilt	2023
14	Washington	2023
15	Arizona St.	2024
16	Arkansas	2024
17	Florida St.	2024
18	Georgia	2024
19	Georgia Tech	2024
20	Iowa	2024
21	LSU	2024
22	Minnesota	2024
23	Mississippi	2024
24	Missouri	2024
25	Oklahoma St.	2024
26	Oregon St.	2024
27	Penn St.	2024
28	Virginia Tech	2024
29	Washington	2024

Cluster 8 − Variance Reduction and Polygon Plots





Cluster 8 − Comparison Table

	Feature	Cluster 8	Non Cluster 8	Variance Reduction	Difference
1	Def_1	−0.05	0.01	0.75	−0.06
2	Def_3	0.87	−0.22	0.75	1.09
3	Def_2	−0.89	0.22	0.73	−1.11
4	def_FTRate	−0.09	0.02	0.70	−0.11
5	def_FG3Rate	0.75	−0.19	0.64	0.93
6	pos_len_def	0.61	−0.15	0.58	0.76
7	def_TOPct	−0.07	0.02	0.56	−0.09
8	def_BlockPct	−0.38	0.09	0.56	−0.48
9	def_eFGPct	−0.06	0.02	0.55	−0.08
10	def_FG2Pct	−0.38	0.09	0.55	−0.48
11	def_NSTRate	0.03	−0.01	0.53	0.03
12	def_ARate	0.27	−0.07	0.33	0.34
13	Exp	−0.41	0.10	0.31	−0.51
14	def_ORPct	−0.49	0.12	0.30	−0.62

Cluster 8 − Tables Summary

	feature	p_value	t_stat
1	Def_1	0.626	−0.49
2	Def_3	0.000	9.96
3	Def_2	0.000	−10.05
4	def_FTRate	0.375	−0.89
5	def_FG3Rate	0.000	7.70
6	pos_len_def	0.000	5.91
7	def_TOPct	0.486	−0.70
8	def_BlockPct	0.000	−3.61
9	def_eFGPct	0.548	−0.60
10	def_FG2Pct	0.001	−3.59
11	def_NSTRate	0.803	0.25
12	def_ARate	0.027	2.26
13	Exp	0.001	−3.33
14	def_ORPct	0.000	−4.06
15	def_eff_adj	0.118	−1.58
16	tempo_adj	0.020	−2.39
17	def_FG3Pct	0.007	2.79
18	def_StlRate	0.045	−2.04
19	Bench	0.522	0.64

	TeamName	Season
1	Boston College	2019
2	Clemson	2019
3	Indiana	2019
4	Oklahoma St.	2019
5	Oregon St.	2019
6	Rutgers	2019
7	Texas	2019
8	UCLA	2019
9	USC	2019
10	California	2020
11	Clemson	2020
12	North Carolina	2020
13	Northwestern	2020
14	Oregon St.	2020
15	Virginia Tech	2020
16	Arizona	2021
17	Indiana	2021
18	Kansas St.	2021
19	Louisville	2021
20	Mississippi St.	2021
21	Nebraska	2021
22	Northwestern	2021
23	Pittsburgh	2021
24	Stanford	2021
25	Boston College	2022
26	Clemson	2022
27	Florida St.	2022
28	Louisville	2022
29	Mississippi St.	2022
30	Northwestern	2022
31	Pittsburgh	2022
32	Boston College	2023
33	Clemson	2023
34	Florida St.	2023
35	Oregon St.	2023
36	Rutgers	2023
37	Stanford	2023
38	Texas Tech	2023
39	Indiana	2024
40	UCLA	2024
41	USC	2024